

 NIET GreaterNoida Estd 2001 Get Future Ready Autonomous Institute	Computer Networks BCSE0502A UNIT-I & 2 Assignment 1	SESSION: 2026-27
Assignment Given Date: 20/08/2026 Assignment Submission: 6/09/2026 Faculty Name: Dr Harsha Gupta/ Mr Prateek Mathur		CLASS/SEM: IT /Vth Maximum Points: 10 Faculty Mail Id: harsha.it@niet.co.in/prateek.mathur@niet.co.in

Note: Write solution of each question in clear handwriting and submit on JUNO.

Q.N	Question Statement	CO
1	Discuss the Goals ,Application and Criteria of computer network in data communication.	K2, CO1
2	Analyze the various topology with diagrams with advantages and disadvantages. If N nodes are in the Network, then find out the no. of connections required for various topology.	K2, CO1
3	Differentiate the OSI and TCP/IP model. Also discuss the Internet model with protocol suit.	K2, CO1
4	Write down the differences between circuit and Packet switching method. Also discuss the Datagram and Virtual circuit switching approach in packet switching.	K2, CO1
5	Sketch the encoding scheme NRZ-L, NRZ-I, Manchester & Differential Manchester encoding for given bit sequence 01011001	K2, CO2
6	Analyze the various types of transmission media in computer networks. Also explain multimode of fiber optics cable with suitable diagrams	K2, CO2
7	Calculate the checksum and check at receiver side for with error or error free for M= 1010110 1110011 1000111	K4, CO2
8	Create hamming code at sender side and calculate checker bits for given data 00110111. Show receiver received single bit error, Find and correct it	K4, CO2
9	Calculate CRC and verify at receiver end for given data 110110 and divisor 1101.	K4, CO2
10	Explain ALOHA and CSMA with suitable diagrams. Also plot graph between Throughput(S) and offered load(G) for pure and slotted ALOHA System.	K4, CO2
11	Analyze the various error control methods(Stop and Wait ARQ & Sliding window ARQ)1 with suitable diagrams	K4 CO2
12	Discuss 1-Persistent, P-Persistent and non-Persistent CSMA with suitable diagrams.	K4, CO2